

LONDON

AC LTD



londonacltd.co.uk
020 3740 4470



Commissioning of an Air- conditioning System

Project Address	

Customer	

System Details	

Test Engineer	Customer
Name:	Name:
Signed:	Signed:
Date:	Date:

	Project:	System:

Report Sheet

Attended site today to commission the air conditioning system.

The system will be put through a full test run checking all indoor units in heating and cooling. Gas charges will be checked with the manufacture guidelines and using virgin refrigerant additionally charge the system if required.

All temperatures will be taken with a temperature thermistor, with the air on temperature reading from the return air grill of the indoor unit and the air off readings taken from the supply air path.

Engineer		Date:	
Comments:			

	Project:	System:
DX Pressure Test Certificate		

Date of Test:	
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System Description:	Full system final test from condenser to HBC Box.
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Condenser Model:		Serial:	
Condenser Model:		Serial:	
Condenser Model:		Serial:	

Method

0.5 Bar	Stage 1	Minimum 3 Minutes
1.5 Bar	Stage 2	Minimum 3 Minutes
3 Bar	Stage 3	Minimum 1 Hour

System Refrigerant:		Tested With:	
Strength Tested To:	BAR	Duration	HRS
Leak Tested To:	BAR	Duration	HRS

I Certify the item Detailed has ben tested in my presence

Engineer Print:		Sign:		Company:	
Witness Print:		Sign:		Company:	
Witness Print:		Sign:		Company:	

	Project:	System:

DX Pressure Test Certificate

Time:

Date:

Pressure

Time:

Date:

Pressure

Engineer		Date:	
Comments:			

	Project:	System:

DX Pressure Test Certificate

Time:

Date:

Pressure

Time:

Date:

Pressure

Engineer		Date:	
Comments:			

	Project:	System:

DX Pressure Test Calibration Certificate

Engineer		Date:	
Comments:			

	Project:	System:
HVRF Water Test Certificate		

Date of Test:	
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System Description:	Full system final test from HBC Box to all evaporator coils.
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Condenser Model:		Serial:	
Condenser Model:		Serial:	
Condenser Model:		Serial:	

Method

0.5 Bar	Stage 1	Minimum 5 Minutes
1.5 Bar	Stage 2	Minimum 5 Minutes
3 Bar	Stage 3	Minimum 1 Hour

		Tested With:	
Leak Tested To:	BAR	Duration	HRS

I Certify the item Detailed has ben tested in my presence

Engineer Print:		Sign:		Company:	
Witness Print:		Sign:		Company:	
Witness Print:		Sign:		Company:	

	Project:	System:

HVRF Water Test Certificate

Time:

Date:

Pressure

Time:

Date:

Pressure

Engineer		Date:	
Comments:			

	Project:	System:

HVRF Water Test Calibration Certificate

Engineer		Date:	
Comments:			

	Project:	System:
Vacuum Test Certificate		

Date of Test:	
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System Description:	Full system final test from condenser to HBC Box.
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Condenser Model:		Serial:	
Condenser Model:		Serial:	
Condenser Model:		Serial:	

Method

Vacuum to 10 TORR	Stage 1	
Break with OFN into the Suction Port to 1 BAR	Stage 2	
Vacuum to 5 TORR (Discharge Valve)	Stage 3	
Evacuate to 2 TORR	Stage 4	Minimum 1 Hour
Rise Test	Stage 5	Minimum 30 Minutes

System Refrigerant:		Tested With:	
Guidelines	TORR	Rise Test Duration	MINS
Vacuum Test	TORR		

I Certify the item Detailed has ben tested in my presence

Engineer Print:		Sign:		Company:	
Witness Print:		Sign:		Company:	
Witness Print:		Sign:		Company:	

	Project:	System:

Vacuum Test Certificate

Time:

Date:

Pressure

Time:

Date:

Pressure

Engineer		Date:	
Comments:			

	Project:	System:
Drain Test Certificate		

Date of Test:	
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System Description:	Water poured into the sump of the pump and allowing the pump to cycle
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Condenser Model:		Serial:	
Condenser Model:		Serial:	
Condenser Model:		Serial:	

Test Type:		Duration:	
Test With:			
Test Amount			

I Certify the item Detailed has ben tested in my presence

Engineer Print:		Sign:		Company:	
Witness Print:		Sign:		Company:	
Witness Print:		Sign:		Company:	



	Contractor and Site Details
Installation Contractor Name and Address	London AC Limited, 1 King William Street, London, EC4N 7AF

hereby confirm that the system detailed in this log book has been pressure tested and evacuated in accordance with the Mitsubishi Electric Method Statements 1 and 2

Site Name and address	
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Note: All sections must be completed for acceptance by MEUK

	System Details
Outdoor Location	
System Reference	
System Model	Select Model
Model Number	Outdoor Unit 1
Unit Serial Number	
Address	
Date	
Commissioning Engineer	

Notes



By Mitsubishi Electric

Please note is important that the correct type and number of Hydro Branch Controllers are selected on this page.

Description:	Model No:	Serial No:	Address:
HBC Controller (Main)	Select Model No.		
HBC Controller (Main 2/Sub 1)	Select Model No.		
HBC Controller (Sub 2)	Select Model No.		
HBC Controller (Sub 3)	Select Model No.		
Hydro Box for Heat Pump	Select Model No.		

R32 HBC Installation Statements must be answered

	YES	NO
Does the HBC/Hydro Box Installation fall in the "NG" zone?		
Have other safety measures been considered?		
"Leak detection Alarm & Mechanical ventilation"		
Or is the room considered as an "Authorised Access Space" (locked & only authorised person like site engineer or manager is allowed)		

[illegible]

Outdoor System Type

Select Model

BC and Sub BC Addition(s)

Piping length outdoor to HBC on HVRF systems

OC to HBC	
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Outdoor Unit Additions

System Indexing

Pipe Work

Diameter	Length Meter	Type	Factor	Calculation
7/8" 22.2mm		High Pressure Gas	kg/m	
3/4" 19.05mm		High Pressure Gas	kg/m	
5/8" 15.88mm		High Pressure Gas	kg/m	
3/4" 19.05mm		Liquid	kg/m	
5/8" 15.88mm		Liquid	kg/m	
1/2" 12.7mm		Liquid	kg/m	
3/8" 9.52mm		Liquid	kg/m	
Total Length	0.00	Meters	Pipe Work Additional	

Charge Calculation Results

Qj Indoor to Outdoor	Valid/Not Valid	Kg's
Qty of Indoor to Outdoor	Valid/Not Valid	
Additional Charge	Kgs	
Total System Charge	Kgs	
Max Addit Charge	Kgs	
Maximum Charge	Kgs	



By Mitsubishi Electric

**Mitsubishi VRF Type Y(S)NW-A OUTDOOR UNIT
OPERATING READINGS USING DIP SWITCH SW4
Please note SW6-10 needs to be set off**

Outdoor Unit 1															
SYMBOL	LOCATION	STANDARD DATA		ACTUAL READING		DIP SWITCH SW4									
	Outdoor Unit 1	Cooling	Heating	Cooling	Heating	1	2	3	4	5	6	7	8	9	0
F	Compressor frequency	Depends on size of unit				ON	☐	☐	☐	☐	☐	☐	☐	☐	☐
						OFF	☐	☐	☐	☐	☐	☐	☐	☐	☐
TH4	Compressor discharge	80 to 95 deg C				ON	☐	☐	☐	☐	☐	☐	☐	☐	☐
						OFF	☐	☐	☐	☐	☐	☐	☐	☐	☐
TH7	Air or Water inlet temperature	35 deg C	7 deg C			ON	☐	☐	☐	☐	☐	☐	☐	☐	☐
						OFF	☐	☐	☐	☐	☐	☐	☐	☐	☐
63LS	Low Pressure	6 to 9 bar				ON	☐	☐	☐	☐	☐	☐	☐	☐	☐
						OFF	☐	☐	☐	☐	☐	☐	☐	☐	☐
63HS1	High Pressure	Faults at 38 to 42 bar				ON	☐	☐	☐	☐	☐	☐	☐	☐	☐
						OFF	☐	☐	☐	☐	☐	☐	☐	☐	☐

To use the dip switches below SW6-10 must be off and SW6-9 must be on.

Main WM Controller						
SYMBOL	LOCATION	STANDARD DATA		READINGS		DIP SWITCH SW4
	WM Controller	Cooling	Heating	Cooling	Heating	
						1 2 3 4 5 6 7 8 9 0
PS 1	Pressure					
TH11	Liquid Side Heating Main HEX.					
TH 13	Gas Side Heating Main HEX.					
TH 12	Liquid Side Cooling Main HEX.					
TH 14	Gas Side Cooling Main HEX.					
PS 3	Pressure					
TH 15	Bypass Inlet Temp.					
TH 16	Bypass Outlet Temp.					

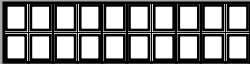
Main WM Controller						
SYMBOL	LOCATION	STANDARD DATA		READINGS		DIP SWITCH SW4
	WM Controller	Cooling	Heating	Cooling	Heating	
						1 2 3 4 5 6 7 8 9 0
PW 1	Pump Inlet Pressure Main Heating HEX.					
PW 2	Pump Outlet Pressure Main Heating HEX.					
TH 32	Water Outlet Temp. Heating Main HEX.					
TH 35	Water Inlet Temp. Heating Main HEX.					
PW 3	Pump Inlet Pressure Main Cooling HEX.					
PW 4	Pump Outlet Pressure Main Cooling HEX.					
TH 33	Water Outlet Temp. Cooling Main HEX.					
TH 34	Water Inlet Temp. Cooling Main HEX.					

Monitor Tool readings for HBC
Not available from outdoor unit dipswitches

Main WM Controller				
SYMBOL	LOCATION	READINGS		DIP SWITCH SW4
		WP Controller	Cooling	Heating
				1 2 3 4 5 6 7 8 9 0
PS 1	Pressure			
TH 11	Liquid Side Heating Main HEX.			
TH 13	Gas Side Heating Main HEX.			
TH 32	Water Outlet Temp. Heating Main HEX.			
TH 12	Liquid Side Cooling Main HEX.			
TH 14	Gas Side Cooling Main HEX.			
TH 33	Water Outlet Temp. Cooling Main HEX.			
TH 15	Bypass Inlet Temp.			
TH 16	Bypass Outlet Temp.			

Main WM Controller				
SYMBOL	LOCATION	READINGS		DIP SWITCH SW4
		WP Controller	Cooling	Heating
				1 2 3 4 5 6 7 8 9 0
PS 1	Pressure			
TH 11	Liquid Side Heating Main HEX.			
TH 13	Gas Side Heating Main HEX.			
TH 32	Water Outlet Temp. Heating Main HEX.			
TH 12	Liquid Side Cooling Main HEX.			
TH 14	Gas Side Cooling Main HEX.			
TH 33	Water Outlet Temp. Cooling Main HEX.			
TH 15	Bypass Inlet Temp.			
TH 16	Bypass Outlet Temp.			

Monitor Tool readings for Hydro Box
Not available from outdoor unit dipswitches

<u>Main WM Controller</u>				
SYMBOL	LOCATION	READINGS		DIP SWITCH SW4
	WP Controller	Cooling	Heating	1 2 3 4 5 6 7 8 9 0
TH11	Refrigerant Temp			
TH13	Refrigerant Temp			
Prf	Refrigerant Pressure			
TH 32	Water Pump Discharge Temp			
TH33	Water Pump Inlet Temp			
PWD	Water Pump Discharge Pressure			
PWS	Water Pump Inlet Pressure			

	Project:	System:

Report Sheet Overview

System has been left in fault free condition.

London AC are Mitsubishi Electric 'Diamond Quality Partners' & Daikin D1+ Partners which means we are regularly audited by third party auditors on behalf of Mitsubishi Electric & Daikin and makes us part of their elite partner schemes.

We can offer extended warranty on all new equipment supplied subject to service and maintenance being undertaken by London AC Ltd in accordance with the manufacturer's recommendations.

For more information please contact;

Tel: 0203740 4470 Option 4
info@lactechservices.co.uk

Engineer		Date:	
Comments:			